

Clinicopathological Characteristics of BRAF V600E Mutated Melanomas in the Dalmatian Region of Croatia.

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A high proportion of cutaneous melanomas harbor activating mutations of the BRAF or NRAS genes, which are components of mitogen-activated protein kinase (MAPK) signal transduction pathway. The importance of BRAF V600E mutation in melanoma is not only related to the possibility of the administration of the targeted therapy, but also to the fact that BRAF V600E mutated melanomas have distinct clinicopathological features. We investigated the clinicopathological features of 80 primary skin melanomas with known BRAF V600E mutation status excised in the Dalmatian region of Croatia, with comparison of these features between the mutated and wild-type group. The frequency of BRAF V600E mutation was 47.5%. In comparison with wild-type melanomas, BRAF V600E mutated melanomas were significantly associated with younger age and female sex ($P=0.014$ and $P=0.011$, respectively). The mutated melanomas were more often located on the extremities, of a nodular type, ulcerated, and with higher median of mitotic index but without significant difference in comparison with wild-type tumors. There were no differences in the depth of invasion and the presence of lymphovascular invasion, tumor infiltrating lymphocytes, and regression between the investigated groups. The frequency of BRAF V600E mutation in our cohort of primary skin melanomas and the clinicopathological features of mutated tumors were similar to those reported in the literature, except for the higher proportion of women observed in our group with mutation.

BRAF V600E mutation as a prognostic factor in cutaneous melanoma patients.

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The prognostic significance of BRAF mutations in the natural course of melanoma is controversial. The aim of study was to assess the prognostic significance of BRAF V600E mutation in cutaneous melanoma patients. A total of 151 melanomas were included in the study. BRAF V600E mutation was detected using the real-time PCR. BRAF V600E mutation rate was 51%. BRAF mutation rate was higher for young patients (61.4%) and upper limbs (63.2%), trunk (59.3%) and head and neck (59.2%) were the most frequently afflicted sites in BRAF-mutant patients, whereas lower limbs were mostly affected in BRAF-wild patients (77.8%). Likewise, acral melanomas rarely harbored BRAF mutation (17.1%). The disease-free survivals regarding the entire and Stage III cohorts were longer in the BRAF-mutant group than in the BRAF-wild group ($p = .006$ and $p = .004$, respectively), whereas Stage I-II patients had no survival differences between BRAF statuses ($p = .2$). Likewise, BRAF-mutant patients had better overall survival (OS) time compared to BRAF-wild patients in all stages ($p = .01$), in Stage III ($p = .01$), and in Stage IV patients ($p = .001$). However, no differences between BRAF statuses were observed in Stage I-II melanomas ($p = .3$). In conclusion, BRAF V600E-mutant melanomas show favorable prognostic impact on both disease-free and OSs in all staged melanomas except local disease.

Therapeutic Strategies in BRAF V600 Wild-Type Cutaneous Melanoma.

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There have been many recent advances in melanoma therapy. While 50% of melanomas have a BRAF

mutation and are a target for BRAF inhibitors, the remaining 50% are BRAF wild-type. Immune checkpoint inhibitors targeting PD-1, cytotoxic T-lymphocyte-associated protein 4 (CTLA4) and lymphocyte activated gene-3 (Lag-3) are all approved for the treatment of patients with advanced BRAF wild-type melanoma; however, treatment of this patient population following initial immune checkpoint blockade is a current therapeutic challenge given the lack of other efficacious options. Here, we briefly review available US FDA-approved therapies for BRAF wild-type melanoma and focus on developing treatment avenues for this heterogeneous group of patients. We review the basics of genomic features of both BRAF mutant and BRAF wild-type melanoma as well as efforts underway to develop new targeted therapies involving the mitogen-activated protein kinase (MAPK) pathway for patients with BRAF wild-type tumors. We then focus on novel immunotherapies, including developing checkpoint inhibitors and agonists, cytokine therapies, oncolytic viruses and tumor-infiltrating lymphocytes, all of which represent potential therapeutic avenues for patients with BRAF wild-type melanoma who progress on currently approved immune checkpoint inhibitors.

Dabrafenib in an elderly patient with metastatic melanoma and BRAF V600R mutation: a case report.

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Approximately 50 % of malignant melanomas harbor activating point mutations in the BRAF gene. Typically, these mutations result in the substitution of the amino acid valine at codon 600 of the gene, and 90-95 % of mutations are either BRAF (V600E) or BRAF (V600K). Specific BRAF inhibitors such as dabrafenib and vemurafenib are the mainstays of treatment in patients with metastatic BRAF-mutant malignant melanomas. The third most common BRAF mutation is V600R, which also leads to increased BRAF signaling. Although evidence exists about the activity of dabrafenib and vemurafenib in patients with the BRAF (V600R) mutation, these patients have been systematically excluded from recent trials with targeted therapies. Here, we report the positive results in terms of survival and quality of life obtained with dabrafenib in an 80-year-old Caucasian male patient with a Charlson Comorbidity Index of 8 diagnosed with metastatic malignant melanoma harboring the BRAF (V600R) mutation. Our patient was treated with dabrafenib for 7 months with minimal toxicity. We also report exploratory analyses of circulating tumor DNA during targeted treatment. Interestingly, the mutation was not detected after starting treatment and became detectable before radiological disease progression. Our report suggests that (1) a relevant benefit can be obtained with a BRAF inhibitor in real-world patients with a malignant melanoma harboring a BRAF (V600R) mutation, and that (2) circulating tumor DNA detection might be of help in assessing tumor burden in everyday clinical practice. The results reported here should encourage the inclusion of patients with BRAF (V600R)-mutated malignant melanomas in future prospective clinical trials with BRAF inhibitors.

BRAF in malignant melanoma progression and metastasis: potentials and challenges.

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Recent advances in gene sequencing have shown that activated BRAF mutations are present in more than 50% of malignant melanomas and contribute to constitutive signals in the MAPK pathway. Besides the importance of its mutations in cell proliferation, BRAF is associated with lymph node, brain and liver

metastasis, along with the loss of PTEN expression and ATG5. Knowledge of this genetic alteration has led to the development of personalized and targeted therapy strategies which block different pathways driving melanoma pathogenesis. Several targeted therapy agents such as vemurafenib, dabrafenib and encorafenib have been approved by the FDA as BRAF inhibitors, as well as other immunotherapies such as anti-CTLA-4 (ipilimumab). However, one of the main challenges is acquired resistance via reactivation of MAPK via CRAF/COT overexpression. Resistance to current BRAF inhibitors is a clinical challenge and one of the strategies to overcome this phenomenon is combination treatment, with the most recently approved combination being BRAF/MEK inhibitors (dabrafenib and trametinib) and BRAF or MEK inhibitors with immuncheckpoint blockers. This review delineates the current role of BRAF in melanoma progression and metastasis. It discusses targeted therapies and resistance mechanisms to BRAF inhibitors, and illustrates strategies to overcome this mechanism with recently approved agents.

Reversal of Resistance in Targeted Therapy of Metastatic Melanoma: Lessons Learned from Vemurafenib (BRAFV600E-Specific Inhibitor).

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Malignant melanoma is the most aggressive form of skin cancer and has a very low survival rate. Over 50% of melanomas harbor various BRAF mutations with the most common being the V600E. BRAFV600E mutation that causes constitutive activation of the MAPK pathway leading to drug-, immune-resistance, apoptosis evasion, proliferation, survival, and metastasis of melanomas. The ATP competitive BRAFV600E selective inhibitor, vemurafenib, has shown dramatic success in clinical trials; promoting tumor regression and an increase in overall survival of patients with metastatic melanoma. Regrettably, vemurafenib-resistance develops over an average of six months, which renders melanomas resistant to other therapeutic strategies. Elucidation of the underlying mechanism(s) of acquisition of vemurafenib-resistance and design of novel approaches to override resistance is the subject of intense clinical and basic research. In this review, we summarize recent developments in therapeutic approaches and clinical investigations on melanomas with BRAFV600E mutation to establish a new platform for the treatment of melanoma.

B-RAF inhibitors: an evolving role in the therapy of malignant melanoma.

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Immunotherapy and chemotherapy benefit few patients with metastatic melanoma, and even fewer experience durable survival benefit. These poor results come from treating melanoma as a single homogeneous disease. Recently, it has been shown that targeting activated tyrosine kinases (oncogenes) can mediate striking clinical benefits in several cancers. In 2002, a mutation at the V600E amino acid of the BRAF serine/threonine kinase was described as present in over 50% of melanomas. The mutation appeared to confer a dependency by the melanoma cancer cell on its activation of the MAP kinase pathway. The frequency and specificity of this mutation (95% at V600E of BRAF) suggests that it may be a potential target for therapy, and recent results with one inhibitor, PLX4032/RG7204, bare this out. This review updates the status of BRAF inhibitors in melanoma and what may be on the horizon.

[Personalized therapy concepts for malignant melanoma].

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Metastatic melanoma is commonly regarded as one of the most difficult tumor entities to treat. Up to 2011 no systemic therapy had been able to achieve a prolongation of overall survival in controlled randomized trials. Cytotoxic chemotherapy resulted in objective remission in only a small subgroup of patients. The growing insight into the molecular pathology and the discovery of frequent mutations made it possible to define melanoma subgroups suitable for targeted therapies. In approximately 50% of melanomas activating mutations of the BRAF gene were identified and can be treated with specific inhibitors. Further mutations which can be approached by targeted therapies are found on the c-Kit and NRAS genes. Another promising approach is immunotherapy aimed to activate cytotoxic T cells. A monoclonal antibody directed against CTLA-4 was approved after convincing results in clinical trials and antibodies against PD-1 or PD-L1 are currently under clinical investigation. Through these achievements life prolonging therapies are available for melanoma patients for the first time.

Targeting BRAF in melanoma: biological and clinical challenges.

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Melanoma is an aggressive form of skin cancer that causes the greatest number of skin cancer-related deaths worldwide. In its early stages malignant melanoma can be cured by surgical resection, but once it has progressed to the metastatic stage it is extremely difficult to treat and does not respond to current therapies. A majority of cutaneous melanomas show activating mutations in the NRAS or BRAF proto-oncogenes, components of the Ras-Raf-Mek-Erk (MAPK) signal transduction pathway. The discovery of activating BRAF mutations in ~50% of all melanomas has proved to be a turning point in the therapeutic management of the disseminated disease. This review summarizes the critical role of BRAF in melanoma pathophysiology, the clinical and pathological determinants of BRAF mutation status and finally addresses the current state of the art of BRAF inhibitors. We further outline the most recent findings on the mechanisms that underlie intrinsic and acquired BRAF inhibitor resistance and describe ongoing preclinical and clinical studies designed to delay or abrogate the onset of therapeutic escape.

Effects of BRAF mutations and BRAF inhibition on immune responses to melanoma.

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Malignant melanoma is associated with poor clinical prognosis; however, novel molecular and immune therapies are now improving patient outcomes. Almost 50% of melanomas harbor targetable activating mutations of BRAF that promote RAS-RAF-MEK-ERK pathway activation and melanoma proliferation. Recent evidence also indicates that melanomas bearing mutant BRAF may also have altered immune responses, suggesting additional avenues for treatment of this patient group. The small molecule inhibitors selective for mutant BRAF induce significant but short-lived clinical responses in a proportion of patients, but also lead to immune stimulatory bystander events, which then subside with the emergence of resistance to inhibition. Simultaneous BRAF and MEK inhibition, and especially combination of BRAF inhibitors with new immunotherapies such as checkpoint blockade antibodies, may further enhance immune activation, or counteract immunosuppressive signals. Preclinical evaluation and ongoing clinical

trials should provide novel insights into the role of immunity in the therapy of BRAF-mutant melanoma.

